Workshop 3: Advanced data wrangling: Extracting ecological signals from noisy systems

3.1 Workshop overview

Welcome to Workshop 3. Today, we are bridging the gap between raw data wrangling and professional, publication-ready data communication.

Today's session is designed to accommodate different paces across the class, ensuring everyone builds a deep, unhurried understanding of master-level coding principles:

1. **The Estuary Data Wrangle Walkthrough** (first ~2 Hours): We will slow down and complete a thorough, step-by-step walkthrough of last week's Estuary Data Rescue Mission. We will focus on identifying all the individual steps required to clean up the raw data and make the final plots.

2. **Exercise 1: Plot Deconstruction (Core Challenge):** You will find, critique, and completely rebuild a published scientific figure using advanced `ggplot2` layers.

3. **Exercise 2: The QFISH Discovery Sprint (Advanced Challenge):** For those moving quickly ahead of the game, this open-ended task requires you to independently source, ingest, wrangle and plot data on Queensland fisheries catch levels.

3.2 Exercise 1: Plot Deconstruction

Data visualisation in marine science is about clear communication. In this exercise, you will critique a piece of published visual data and then wrangle and re-plot your own alternative.

While you are encouraged to work in pairs or small groups to hunt for your figures and critique any shortcomings or areas for improvement, you generate and execute your code **individually** within your own local repository, so that you can embed directly into your own ePortfolio.

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Part 1. Find your data.

Find a plot online, either in a scientific journal, a newspaper, or some other place (e.g. the world bank data portal) which:

- Interests you and your partner.

- You can improve on the plot.
- The data in the plot are complex enough for you to create an interesting graphic. We recommend at least two variables, perhaps more.
- Post the URL of your chart to our subject Google Doc. Make sure no-one else has chosen the same plot as you.

Part 2: Visual Critique

Take a high-resolution screenshot of the original plot. In your notebook or a rough draft document, explicitly define the following items:

- **The Message:** What core story or ecological relationship were the authors attempting to show?
- **The Flaws:** Identify at least two significant data presentation issues. Look for misleading geometric choices (e.g., bar charts representing continuous means), inaccessible or unprincipled colour palettes, omitted measurement units, hidden data distributions, or cluttered layout elements.

Part 3: Plot deconstruction

To reconstruct the plot, you need access to the underlying data. You can acquire these via two paths:

- **The Source Document:** Dig into the paper's supplementary materials, data appendices, or public repositories (like Dryad or Figshare) to see if you can access and download the source file.
- **Manual Digitisation:** If the raw data are unavailable, there are digital data harvesting tools available to you, such as PlotDigitizer. This browser application allows you to upload an image of the chart, calibrate the X/Y axes, and manually click the data points to compile and export a clean .csv file.

Part 4: Plot reconstruction

Open your RStudio Project and initialize a new Quarto (.qmd) document. Your task is to completely recreate a superior version of the plot:

- Import your extracted dataset.
- Fix any problems or flaws you identified in Part 2.
- Step away from the authors' original geometry if necessary (e.g., map raw points using `geom_jitter()` or `geom_boxplot()` instead of relying on standard dynamited bar charts).
- Apply professional layout standards: select an accessible palette (such as those found in the viridis package), remove default grey backgrounds by implementing theme_minimal()

or theme_classic(), and write a comprehensive, publication-grade figure caption.

Your rendered HTML file, showing both the original screenshot, your critique, and your new, high-fidelity plot, will form a core piece of your ePortfolio.

3.3 Exercise: Queensland fisheries data discovery challenge

If you have completed your Plot Deconstruction and want to push your data wrangling and plotting skills, the training wheels are coming off. This task requires you to source unstructured open data from a live government interface, establish your own data pipeline, and build a management report from scratch. Here, we will use real-world marine data collected as part of Queensland fisheries QFISH database to develop a report, suitable for reporting to managers, community groups or for your own research.

Part 1: Sourcing data

Navigate to the Queensland Fisheries QFISH Portal.

- Explore the available interactive data tabs (for example, the *Shark Control Program* database or the *Commercial Catch Logbooks*).
- Configure a query that incorporates catch abundance data distributed over both space (e.g., regional management zones) and time (calendar years) for specific species groups.
- Export the dataset as a raw .csv table and securely move it into the data/ subdirectory of your local project environment.

Part 2: Independent Wrangling & Communication

Create a new Quarto document and build an independent data pipeline that performs the following steps:

1. **Data loading and wrangling:** Government data tables frequently export with complex metadata headers, unformatted text strings, and trailing summary rows. Use your growing data wrangling skills to tidy the data into usable format.
2. **Ecological Aggregation:** Group and summarise the catch timelines to expose an underlying trend or answer a specific management question (e.g., tracking the long-term temporal shifts in catch rates across different regions or understanding by-catch).
3. **Visual Layout:** Build a highly polished line or scatter time-series. Ensure all variable names are written out fully, legends are clean, and a custom ggplot2 theme is implemented.

Part 3: Report writing

In your Quarto file, add clear Markdown paragraph blocks that contextualise your plot. Frame this document as a professional briefing note intended for fisheries managers or a regional community advisory group.

Render your file to an HTML report and push the update to your GitHub portfolio. In your ePortfolio, your report should contain details of the wrangling steps, where you got the data, and the analysis you developed (follow good coding style!) . Finally, include your plot together with a detailed caption of what your plot shows. Your final ePortfolio should be suitable for sharing with a colleague or the public, so that they can understand the dataset you have presented to them.